

i have to do EDA on certain datasets -- all of them are supervised and some are regression, some are classification..so we have to develop eda fns that takes any dataset of a particular type and performs eda--like a general one for regression, general for classification.. the eda pipeline/process differs for regression type dataset and classification type dataset right

Yes. The **overall EDA workflow is similar**, but **some stages differ depending on whether the problem is regression or classification**, mainly because of the **target variable**.

Think of it this way:

- **80–90% of EDA is common** to every supervised dataset.
- The remaining **10–20% depends on the target type**.

Common EDA Pipeline (Both Regression & Classification)

These stages can be exactly the same.

Stage 1 — Basic Dataset Information

Questions:

- Number of rows?
- Number of columns?
- Feature names?
- Data types?
- Memory usage?

Functions

Python



Run

```
df.shape
df.info()
df.dtypes
```

Stage 2 — Data Quality Check

Questions:

- Missing values?
- Duplicate rows?
- Wrong datatypes?
- Constant columns?
- High-cardinality columns?

Functions

Python



Run

```
df.isnull().sum()  
df.duplicated().sum()  
df.nunique()
```

Stage 3 — Univariate Analysis

Study every feature individually.

Numerical

- Histogram
- KDE
- Boxplot
- Mean
- Median
- Std
- Skewness

Categorical

- Countplot
- Frequency
- Percentage

Stage 4 — Outlier Detection

Numerical columns

- Boxplot
- IQR
- Z-score

Stage 5 — Feature Relationships

Among independent variables

Numerical vs Numerical

- Correlation Matrix
- Heatmap
- Pairplot

Categorical vs Numerical

- Boxplot
- Violinplot

Categorical vs Categorical

- Crosstab
- Chi-square

Now the pipeline changes...

Regression EDA

Target = Continuous

Examples

LoanAmount

HousePrice

Salary

Temperature

Stage 6 — Target Analysis

Questions

How is target distributed?

Normal?

Skewed?

Outliers?

Plots

Histogram

KDE

Boxplot

Statistics

Mean

Median

Std

Skewness

Stage 7 — Feature vs Target

This is the most important stage.

For every feature compare with target.

Numerical → Target

Scatterplot

Regression Line

Correlation

Questions

Positive relation?

Negative?

Linear?

Nonlinear?

Outliers?

Categorical → Target

Boxplot

Violinplot

Questions

Does target differ between categories?

Example

Education vs Salary

Stage 8 — Multicollinearity

Correlation Matrix

VIF

Purpose

Detect redundant predictors.

Classification EDA

Target = Categories

Examples

Spam / Ham

Yes / No

0 / 1

Species

Stage 6 — Target Variable Analysis

This stage is completely different.

Questions

How many classes?

Balanced?

Imbalanced?

Functions

Python



Run

```
df[target].value_counts()  
df[target].value_counts(normalize=True)
```

Plots

Countplot

Pie Chart (optional)

Decision

Balanced



Accuracy can be a useful metric.

Imbalanced



Need metrics like

- Precision
- Recall
- F1-score
- ROC-AUC
- Stratified train/test split

Stage 7 — Feature vs Target

Again differs.

Numerical → Target

Plots

Boxplot

Violinplot

Stripplot

Questions

Do feature values differ across classes?

Example

Age vs Purchased

Categorical → Target

Plots

Countplot (with hue)

Stacked Bar

Crosstab

Questions

Is category associated with target?

Example

Gender vs Purchased

Statistical Tests

Chi-square

ANOVA (if applicable)

Stage 8 — Feature Importance (Optional)

Questions

Which variables separate classes?

Methods

- Mutual Information
- Information Gain
- Random Forest Feature Importance
- Logistic Regression Coefficients

Comparison

Stage	Regression	Classification
Basic Info	✓	✓
Missing Values	✓	✓
Duplicates	✓	✓
Data Types	✓	✓
Univariate Analysis	✓	✓
Outliers	✓	✓
Correlation Between Features	✓	✓
Target Analysis	Distribution, skewness, outliers	Class balance, class counts
Feature vs Target	Scatterplots, correlation	Boxplots, countplots, crosstabs
Statistical Tests	Pearson, Spearman, VIF	Chi-square, ANOVA, Mutual Information
Main Goal	Predict a continuous value	Separate classes

A Better Software Design

Since you're building **reusable EDA functions**, avoid creating two completely separate pipelines. Instead, create:

EDA/



```
|
|— common_eda(df)
|   |— dataset_info()
|   |— data_quality()
|   |— missing_values()
|   |— duplicates()
|   |— datatype_summary()
|   |— univariate_analysis()
|   |— outlier_analysis()
|   |— feature_relationships()
|
|— regression_eda(df, target)
|   |— target_distribution()
|   |— numerical_vs_target()
|   |— categorical_vs_target()
|   |— multicollinearity()
|
|— classification_eda(df, target)
|   |— class_balance()
|   |— numerical_vs_target()
|   |— categorical_vs_target()
|   |— class_association_tests()
```

This design minimizes duplicated code because all shared steps live in `common_eda()`, while only the target-specific analyses differ. It also makes the pipeline easier to maintain and extend (for example, adding a future `time_series_eda()` without rewriting the common stages).

where to study EDA and I want to learn all concepts --say if data is right skewed but u can't get rid off outliers so we do log transformation to reduce skewness..so that mean is not affected..I wnt to cover all the subtopics like this so that I can do the coding in pythn

You're thinking about EDA the right way. The goal is **not** to memorize Python functions like `sns.histplot()` or `df.describe()`, but to understand:

"What am I checking? Why am I checking it? What does each result imply? What should I do next?"

For example:

- Right-skewed distribution → Mean > Median

- Why is that bad? → Many ML models assume symmetric distributions or are affected by extreme values.
- Can I remove outliers? → If they're valid observations, maybe not.
- Then what? → Log transform, Box-Cox, Yeo-Johnson, RobustScaler, or use tree-based models depending on the situation.

This is the level of understanding that makes you good at EDA.

I recommend learning EDA in this order

Phase 1: Statistics (the foundation)

Without statistics, EDA becomes just plotting graphs.

Cover these topics thoroughly:

Descriptive Statistics

- Mean
- Median
- Mode
- Range
- Variance
- Standard Deviation
- Percentiles
- Quartiles
- IQR
- Five-number summary

Ask yourself for each:

- What does it measure?
 - When is it useful?
 - When does it fail?
-

Distributions

Learn:

- Normal distribution
- Uniform distribution
- Right skew
- Left skew

- Multimodal
- Bimodal

Then ask:

- Why is normality important?
 - Which algorithms require it?
 - What if data isn't normal?
-

Skewness

Understand:

- Positive skew
- Negative skew
- Causes
- Effects

Then learn:

When should we:

- Ignore it?
 - Log transform?
 - Square root transform?
 - Box-Cox transform?
 - Yeo-Johnson transform?
-

Outliers

Don't just learn IQR.

Learn

Detection

- IQR
- Z-score
- Modified Z-score
- Isolation Forest
- Local Outlier Factor

Then ask

When should we

- Remove?
 - Keep?
 - Cap (Winsorize)?
 - Transform?
 - Use robust models?
-

Correlation

Learn

- Pearson
- Spearman
- Kendall

Know

When each should be used.

Covariance

Understand how it's different from correlation.

Categorical Statistics

Learn

- Frequency
 - Relative frequency
 - Contingency tables
 - Chi-square test
-

Phase 2: Data Cleaning

Understand why each issue matters and how to handle it.

Missing Values

Study:

- MCAR
- MAR
- MNAR

Methods

- Delete rows
- Delete columns
- Mean imputation
- Median imputation
- Mode imputation
- KNN imputation
- Regression imputation
- MICE

For every method learn:

- Advantages
 - Disadvantages
 - Assumptions
 - When to use
 - When not to use
-

Duplicates

- Exact duplicates
 - Near duplicates
 - Intentional duplicates
-

Incorrect Data Types

Examples

Age stored as string

Date stored as object

Zip codes treated as integers

Inconsistent Data

Examples

Male

male

MALE

Feature Scaling

Understand

- StandardScaler
- MinMaxScaler
- RobustScaler
- MaxAbsScaler

For each:

- Formula
 - Effect
 - Suitable algorithms
 - Unsuitable algorithms
-

Encoding

- One-hot encoding
- Label encoding
- Ordinal encoding
- Target encoding
- Frequency encoding

Know when each is appropriate.

Phase 3: Visualization

Don't just learn how to create plots—learn what questions they answer.

For every plot ask:

- What is it used for?
- When should I use it?
- What patterns can it reveal?
- When is it misleading?

Examples:

Plot	Purpose
Histogram	Distribution

Plot	Purpose
KDE	Smoothed distribution
Boxplot	Outliers, spread
Violin plot	Distribution + density
Scatter plot	Relationship
Pairplot	Pairwise relationships
Heatmap	Correlation
Countplot	Category frequencies
Bar plot	Aggregated values
Line plot	Trends
ECDF	Distribution comparison

Phase 4: Feature Engineering Concepts

Learn why we create new features.

Topics include:

- Log transformation
- Polynomial features
- Interaction features
- Binning
- Date/time feature extraction
- Cyclical encoding
- Feature selection
- Feature extraction

Phase 5: Regression-specific EDA

Study:

- Target distribution

- Residuals (after modeling)
 - Heteroscedasticity
 - Linearity
 - Multicollinearity
 - Variance Inflation Factor (VIF)
-

Phase 6: Classification-specific EDA

Study:

- Class imbalance
 - Stratified sampling
 - Confusion matrix
 - Precision
 - Recall
 - F1-score
 - ROC-AUC
 - PR-AUC
 - Decision thresholds
-

Phase 7: Statistical Tests

Know the assumptions, not just the names.

- Shapiro-Wilk (normality)
 - Kolmogorov-Smirnov (distribution comparison)
 - Levene's test (equal variances)
 - t-test
 - ANOVA
 - Mann-Whitney U
 - Chi-square
 - Pearson correlation test
 - Spearman correlation test
-

Best Resources

1. Books (best for deep understanding)

If you want strong conceptual knowledge:

- **Practical Statistics for Data Scientists** (Bruce & Bruce) — excellent bridge between statistics and data science.
 - **Think Stats** (Allen B. Downey) — beginner-friendly with intuition and Python examples.
 - **An Introduction to Statistical Learning (ISLR)** — connects statistics to machine learning.
-

2. Documentation

Read the guides for:

- pandas
- matplotlib
- scikit-learn

These teach the tools, but pair them with statistical understanding.

3. Practice

The fastest way to internalize EDA is to repeat the process on many datasets:

- Titanic (classification)
 - Loan Prediction (classification)
 - House Prices (regression)
 - California Housing (regression)
 - Wine Quality (classification or regression)
 - Heart Disease (classification)
 - Email Spam (classification)
 - MNIST (classification)
-

A roadmap for mastering EDA

1. Statistics fundamentals
2. Data cleaning and preprocessing
3. Visualization and interpretation
4. Feature engineering concepts
5. Regression-specific EDA
6. Classification-specific EDA
7. Statistical tests and assumptions
8. Practice on diverse real-world datasets

At every step, follow this mindset:

Observation → Interpretation → Decision → Action

For example:

- **Observation:** Income is highly right-skewed.
- **Interpretation:** A few very large incomes are pulling the mean upward.
- **Decision:** The outliers are legitimate, so don't remove them.
- **Action:** Apply a log transformation or use a robust model/statistic if appropriate.

That reasoning process is what separates simply "doing EDA" from using EDA to make informed preprocessing and modeling decisions.

Monday 6:28 PM

Normal distribution
Uniform distribution
Right skew
Left skew
Multimodal
Bimodal

are these distributions enough to learn? or should i see
poison etc and?

For **EDA and Machine Learning**, those six are **NOT** enough, but you also **don't need to learn every probability distribution** (there are 100+).

Instead, learn the distributions that you will actually encounter in data science.

Priority 1 (Must Know ★★★★★)

These are essential for EDA.

1. Normal (Gaussian) ★★★★★

- Bell curve
- Most statistical tests assume this
- Used in linear regression assumptions
- Standardization, z-scores

2. Uniform

- Every value is equally likely
- Useful for understanding randomness

3. Right (Positive) Skew

- Income
- House prices
- Transaction amounts
- Population of cities

4. Left (Negative) Skew

- Easy exam scores
- Age at retirement

5. Bimodal

- Two peaks
- Often indicates two underlying groups

6. Multimodal

- Three or more peaks
- Often suggests mixed populations or hidden categories

These six cover **90% of what you'll encounter during EDA**.

Priority 2 (Should Know ★★★★★)

These aren't common in raw EDA, but they're very important in machine learning and statistics.

7. Poisson Distribution

Use when counting events.

Examples:

- Emails received per hour
- Customers entering a shop
- Calls to a call center
- Website visits per minute

Questions to learn:

- When is Poisson appropriate?
 - Difference between Poisson and Normal?
 - Why does Poisson become approximately Normal when λ is large?
-

8. Binomial Distribution

Very important for classification.

Examples:

- Spam or not spam
- Pass or fail
- Customer buys or not
- Defective product or not

You'll encounter this in:

- Logistic Regression
 - Probability
 - A/B testing
-

9. Exponential Distribution

Models **time until the next event**.

Examples:

- Time until next customer arrives
 - Time until a machine fails
 - Waiting time between emails
-

Priority 3 (Know the Idea ★★☆☆)

You don't need to master these initially, but recognize where they're used.

Gamma Distribution

- Insurance claims
- Waiting times
- Reliability analysis

Beta Distribution

- Probabilities between 0 and 1
- Bayesian statistics
- Conversion rates

Log-Normal Distribution ★★★★★

This one is surprisingly useful.

If

$\log(X)$



is normally distributed,

then

X



is log-normal.

Examples:

- Income
- Stock prices
- House prices
- File sizes

This directly connects to why we use **log transformations** in EDA.

Priority 4 (Just Know They Exist)

You can safely postpone these until later.

- Weibull
- Geometric
- Negative Binomial
- Student's t
- Chi-square
- F-distribution
- Cauchy
- Laplace

These are mostly encountered in advanced statistics, survival analysis, or specific hypothesis tests rather than everyday EDA.

For an EDA expert, don't just learn the names

For **every** distribution, answer these questions:

1. What does it look like?
2. Where does it occur in real life?
3. How can I identify it from a histogram or KDE plot?
4. What does it imply about the data?

5. What problems can it cause?
6. Which ML algorithms are affected?
7. Should I transform the data?
8. If yes, which transformation should I use?
9. Which statistical tests assume this distribution?

For example:

Distribution	Real-life examples	Typical action in EDA
Normal	Heights, measurement errors	Usually no transformation needed
Right-skewed	Income, prices	Consider log, Box-Cox, or Yeo-Johnson if appropriate
Left-skewed	Easy exam scores	Reflect + transform if necessary (less common)
Uniform	Random numbers	Usually leave unchanged
Bimodal	Male & female heights combined	Investigate hidden groups or segment the data
Multimodal	Mixed customer populations	Look for clusters or subgroup effects
Poisson	Number of calls, emails	Common for count data; don't force normality unnecessarily
Binomial	Yes/No outcomes	Typical target distribution in classification
Exponential	Waiting times	Often right-skewed; model accordingly
Log-normal	Income, house prices	Log transformation is often useful

My recommended learning order

1.  Normal
2.  Uniform
3.  Right-skew
4.  Left-skew
5.  Bimodal
6.  Multimodal
7.  Log-normal
8.  Binomial

9. ✓ Poisson
10. ✓ Exponential

That sequence gives you nearly everything you need for practical EDA, preprocessing, and the statistical assumptions behind common machine learning algorithms.

should I learn these with a visualisation? like histograms or smthg like ?

Absolutely. In fact, I would say don't learn distributions without visualizing them.

EDA is about **looking at data**. When you see a histogram or KDE plot, you should be able to immediately think:

"This is right-skewed."

"There are two peaks, so it's bimodal."

"This looks approximately normal."

"The tails are heavy; there may be outliers."

That's the skill you'll actually use.

Here's how you should study each distribution

For every distribution, make notes under these headings:

1. Shape (Visual)

Look at:

- Histogram
- KDE plot
- Boxplot

Ask:

- What does it look like?
- Is it symmetric?
- Where is the tail?
- Are there multiple peaks?

2. Real-life examples

Example:

Right-skewed

- Salary
- House prices
- Website traffic
- Transaction amounts

This helps you recognize patterns in real datasets.

3. Properties

Example for Normal Distribution:

- Mean = Median = Mode
 - Symmetric
 - Bell-shaped
 - No skewness
 - About 68–95–99.7% of data lies within 1, 2, and 3 standard deviations
-

4. How to identify it in Python

Python



Run

```
sns.histplot(df["Income"], kde=True)
```

```
sns.boxplot(x=df["Income"])
```

```
df["Income"].skew()
```

5. What problems does it cause?

Example:

Right-skewed

Problems:

- Mean gets pulled upward
- Standard deviation increases

- Some statistical tests become less reliable
 - Linear regression assumptions may be affected
-

6. How to handle it

Example:

If outliers are genuine:

- ✓ Log transform
 - ✓ Box-Cox
 - ✓ Yeo-Johnson
 - ✓ Use robust statistics (median, IQR)
 - ✓ Consider tree-based models if transformation isn't appropriate
-

7. When should you leave it alone?

This is often overlooked.

For example:

If you're using

- Random Forest
- Decision Tree
- XGBoost

you often **don't need to normalize or transform** a skewed feature because these models are generally insensitive to monotonic transformations.

One more thing to learn

Don't just memorize the names—learn to **recognize them instantly**.

If I showed you six histograms, you should be able to identify them in seconds.

For example:

Normal



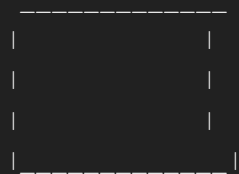
Right Skew



Left Skew



Uniform



Bimodal



Multimodal



My recommendation for mastering EDA

For each distribution, follow this sequence:

1. **Understand the concept** (5–10 minutes).
2. **Look at multiple histograms/KDE plots** until you can recognize the shape instantly.
3. **Generate synthetic data in Python** using NumPy or SciPy and visualize it yourself.
4. **Find a real dataset** where that distribution appears.
5. **Practice deciding what, if anything, should be done** (transform, keep as is, investigate, etc.).

This approach builds the intuition you'll use in real EDA. When you later inspect an unfamiliar dataset, you'll naturally connect the plot you see to the appropriate preprocessing or modeling decisions instead of just recognizing a name.

Yesterday 5:07 PM

Page 4: Non-Technical Events (nontechnical.html)

Purpose

Display entertainment and fun events.

Content

Event 1: Photography

Event 2: Quiz Competition

Event 3: Treasure Hunt

Event 4: Gaming Tournament

For each event:

- Description
- Rules

Show more ▾

Here's content you can use for the **Non-Technical Events** page of **Technova '26**.

 Edit



Non-Technical Events

Take a break from coding and enjoy a series of exciting non-technical events designed to test your creativity, teamwork, and problem-solving skills. Whether you're a photographer, gamer, quiz enthusiast, or adventure seeker, there's something for everyone at Technova '26.

Event 1: Photography Contest

Description:

Capture moments that tell a story! Participants can showcase their photography skills by submitting original photographs based on the event theme.

Rules:

- Individual participation only.
- Photos must be original and unedited except for basic color correction.
- Images should be submitted in JPEG or PNG format.
- Any form of plagiarism will lead to disqualification.

Registration Fee: ₹100 per participant

Prize:

- 🏆 First Prize: ₹3,000
 - 🥈 Second Prize: ₹1,500
 - Certificates for all participants
-

Event 2: Quiz Competition

Description:

Test your knowledge across technology, current affairs, science, entertainment, and general knowledge in an exciting quiz competition.

Rules:

- Team of 2 members.
- Multiple rounds including MCQs, visual identification, and rapid-fire.
- Use of mobile phones or external assistance is strictly prohibited.
- Quiz master's decision will be final.

Registration Fee: ₹150 per team

Prize:

- 🏆 First Prize: ₹4,000
 - 🥈 Second Prize: ₹2,000
 - Participation certificates
-

Event 3: Treasure Hunt

Description:

Follow clues, solve puzzles, and race against other teams to uncover the hidden treasure around the campus.

Rules:

- Team of 3–4 members.
- Teams must follow the designated campus boundaries.
- Damaging property or disturbing others will result in disqualification.
- The team that finds the treasure first wins.

Registration Fee: ₹200 per team

Prize:

- 🏆 Winning Team: ₹5,000
 - Runner-up Team: ₹2,500
 - Certificates for all participants
-

Event 4: Gaming Tournament

Description:

Compete against fellow gamers in an action-packed gaming tournament featuring popular multiplayer games.

Rules:

- Individual or team participation based on the selected game.
- Fair play is mandatory; cheating or exploiting bugs is prohibited.
- Participants must report at least 15 minutes before the match.
- Tournament format and fixtures will be announced on the event day.

Registration Fee: ₹200 per participant/team

Prize:

- 🏆 First Prize: ₹6,000
- 🥈 Second Prize: ₹3,000
- Participation certificates

You can easily style this using cards or sections in your HTML page, with each event displayed in its own card containing the description, rules, registration fee, and prize details.